

Deep Learning for Wine Quality Prediction: Neural Network Optimization and Performance Analysis

By Aaron Niecestro

Project Objective

What is Deep Learning?

- Deep Learning is a subset of machine learning that uses artificial neural networks with multiple hidden layers.
- These networks learn complex nonlinear relationships directly from data.
- Deep learning models automatically discover patterns and interactions that may be difficult to identify using traditional statistical approaches.

Goal

Develop and optimize neural network models capable of predicting wine quality from physicochemical measurements.

Objectives

- Establish a baseline neural network model.
- Evaluate the impact of feature scaling techniques.
- Compare activation functions.
- Compare neural network architectures.
- Optimize epochs, optimizers, and regularization methods.
- Identify the strongest overall model configuration.

Business Relevance

Understanding how wine chemistry influences quality can help support:

- Quality assessment
- Production consistency
- Process optimization
- Data-driven decision making

Why It Matters

Wine quality is influenced by complex interactions among acidity, alcohol, sulfur compounds, density, and other chemical properties.

Deep learning provides a powerful framework for learning these relationships and generating accurate quality predictions

Dataset Description

UCI Wine Quality Dataset

- Physicochemical measurements collected from Portuguese Vinho Verde wines.

Dataset Size

- 5,320 observations
- 11 chemistry features
- Quality scores from 3 to 9

Input Features

- Acidity measurements
- Sugar content
- Sulfur dioxide levels
- Density
- pH
- Alcohol concentration

Target Variable

- Wine Quality
- Predicted using neural networks

Interpretation

The dataset contains numerical chemistry measurements describing the physical and chemical composition of both red and white wines. These measurements serve as inputs to the neural network, while quality scores act as the prediction target.

Variable	Description	Was this Feature used for Modeling?
Fixed Acidity	Tartaric acid content	Yes
Volatile Acidity	Acetic acid concentration	Yes
Citric Acid	Citric acid level	Yes
Residual Sugar	Remaining sugar	Yes
Chlorides	Salt concentration	Yes
Free Sulfur Dioxide	SO_2 level	Yes
Total Sulfur Dioxide	Total SO_2	Yes
Density	Wine density	Yes
pH	Acidity measure	Yes
Sulphates	Sulfate concentration	Yes
Alcohol	Alcohol percentage	Yes
Quality	Wine quality score	Used for interpretation and evaluation
Type	Red or White	Used for interpretation and evaluation

Brief View of Wine Data

Variable	Mean	Range
Fixed Acidity (g/dm^3)	7.22	3.8-15.9
Volatile Acidity (g/dm^3)	0.34	0.08-1.58
Residual Sugar (g/dm^3)	5.05	0.6-65.8
Total Sulfur Dioxide (mg/dm^3)	114.11	6-440
Density (g/cm^3)	0.9945	0.987-1.039
pH	3.22	2.72-4.01
Alcohol (% Vol.)	10.55	8.0-14.9

Metric	Value
Observations	5,320
Features	11 Chemistry Variables
Wine Types	Red & White
Quality Range	3-9
Avg Alcohol	10.55%
Avg Residual Sugar	5.05 g/dm^3
Avg pH	3.22
Avg Total Sulfur Dioxide	114.11 mg/dm^3

Wine Quality Distribution by Wine Type

Key Findings

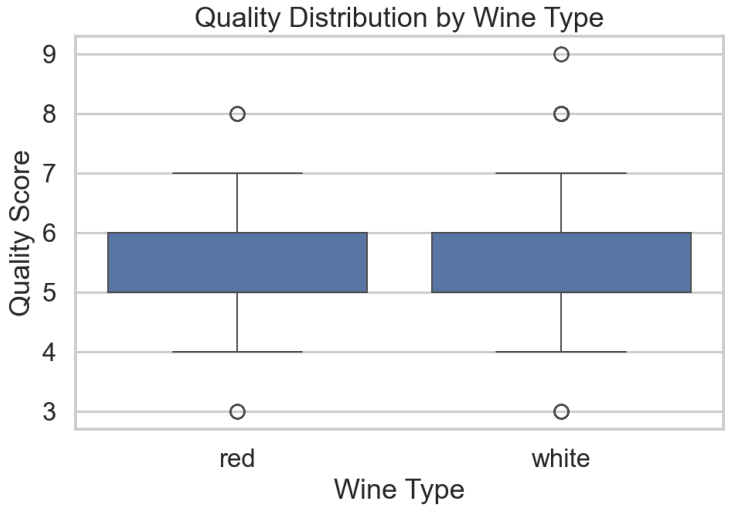
- Alcohol exhibited the strongest positive relationship with wine quality.
- Volatile acidity showed a negative relationship with quality.
- Several chemistry variables demonstrated meaningful correlations with the target variable.
- Most wines were assigned quality scores between 5 and 6.

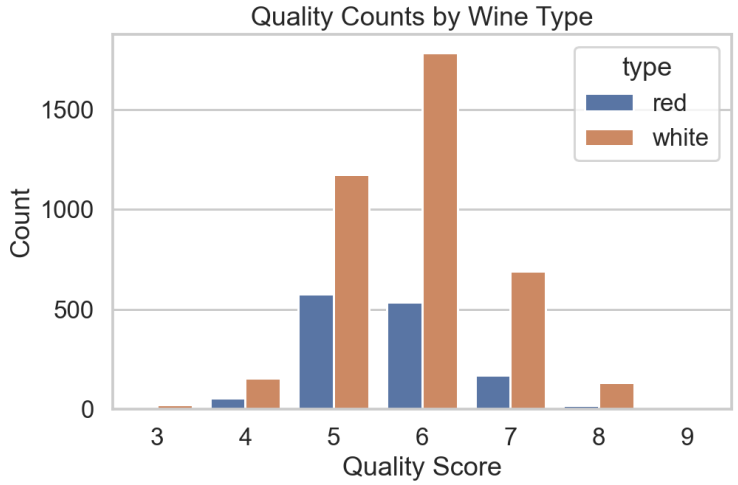
Interpretation

The observed relationships indicate that wine chemistry contains predictive information that can be leveraged by deep learning models.

Takeaway

Meaningful relationships exist between chemistry measurements and wine quality, supporting the use of neural networks for predictive modeling.





Data Preparation & Analysis Approach

Data Preparation

- Combined red and white wine datasets
- Removed duplicate observations
- Selected physicochemical predictors
- Defined wine quality as the target variable

Modeling Workflow

- Train/Test Split (80/20)
- Feature Scaling Analysis
- Activation Function Analysis
- Architecture Comparison
- Epoch Optimization
- Optimizer Benchmarking
- Regularization Analysis
- Final Model Selection

Why This Matters

- Neural networks are highly sensitive to feature scaling and architecture design.
- A structured optimization workflow helps identify the most effective model configuration.

Neural Networks Explained

What Is a Neural Network?

- A machine learning model inspired by biological neurons.
- Learns relationships between input features and target outcomes.
- Uses hidden layers to capture nonlinear patterns.

Why Use Deep Learning?

- Captures complex feature interactions.
- Learns nonlinear relationships.
- Adapts well to regression problems.

Evaluation Metrics

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- Adjusted R^2
- Learning Curves
  - Training Loss
  - Validation Loss

Key Question

- Which neural network configuration produces the strongest wine quality predictions?

Model Evaluation Metrics

Prediction Accuracy	Model Performance	Training Behavior	Why These Metrics Matter
• MAE → Average prediction error • RMSE → Error with emphasis on larger mistakes	• Adjusted R^2 → Explanatory power • Higher values indicate a better fit	• Training Loss → Learning progress • Validation Loss → Generalization performance	• MAE & RMSE measure accuracy • Adjusted R^2 measures model fit • Loss metrics measure stability and generalization

Baseline Neural Network Model Selection

Objective	Perceptron	Multi-Layer Perceptron (MLP)	Why Compare Them?	Key Question
Determine whether a Perceptron or Multi-Layer Perceptron (MLP) provides stronger predictive performance for wine quality prediction.	• Single-layer neural network • Learns linear relationships • Simplest neural network architecture • Used as the baseline model	• Contains one or more hidden layers • Learns complex nonlinear relationships • Greater modeling flexibility • Better suited for predictive modeling tasks	• Determine whether additional network complexity improves prediction accuracy. • Establish the most appropriate neural network foundation before optimization.	Does the added complexity of an MLP improve prediction performance enough to justify its use?

Initial Model Performance

Findings

- MLP achieved lower MAE and RMSE than the Perceptron.
- MLP produced a higher Adjusted R^2 (0.2924 vs. 0.2295).
- Hidden layers improved the model's predictive capability.

Interpretation

- Wine quality depends on nonlinear relationships that cannot be fully captured by a single-layer Perceptron.
- The MLP was better able to learn complex interactions among wine chemistry variables.

Conclusion

- **Winner: Multi-Layer Perceptron (MLP)**
- Lower Prediction Error
- Higher Adjusted R^2
- Stronger Predictive Performance

Model	MAE	RMSE	Adjusted R^2
Perceptron	0.5894	0.7686	0.2295
MLP	0.5660	0.7365	0.2924

Feature Scaling Comparison

Objective	No Scaling	StandardScaler	MinMaxScaler	Why Compare?
Determine how feature scaling influences neural network performance, convergence behavior, and predictive accuracy.	• Uses raw feature values. • Preserves original measurement units. • Serves as the baseline preprocessing approach.	• Centers data around a mean of zero. • Scales features using standard deviation. • Commonly used with neural networks.	• Scales all features into the range of 0 to 1. • Preserves relative relationships between observations. • Frequently used when features have very different ranges.	• Neural networks use gradient-based optimization. • Large differences in feature magnitudes can negatively affect learning efficiency, convergence speed, and overall model performance.

Feature Scaling Comparison Plots

Findings

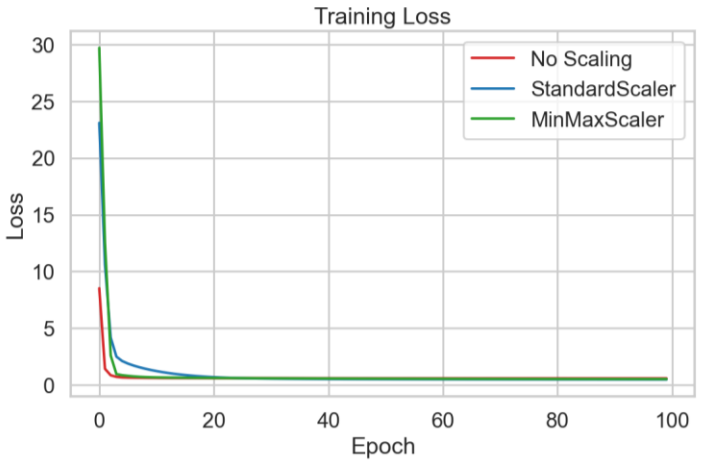
- StandardScaler achieved the best overall performance.
- MinMaxScaler produced similar but slightly weaker results.
- No Scaling resulted in higher error and lower Adjusted R^2.

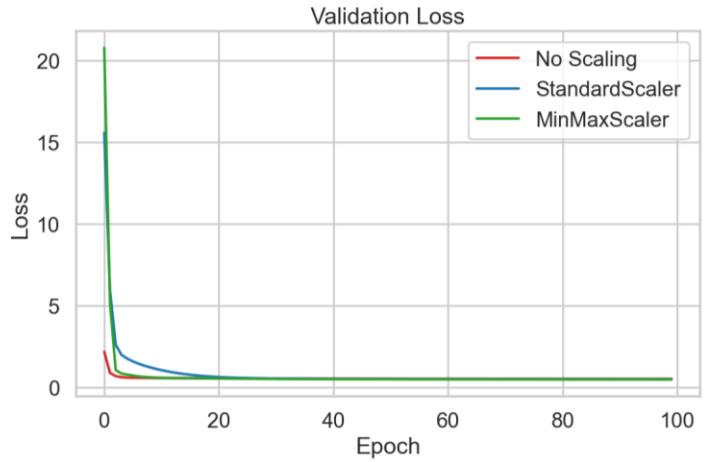
Interpretation

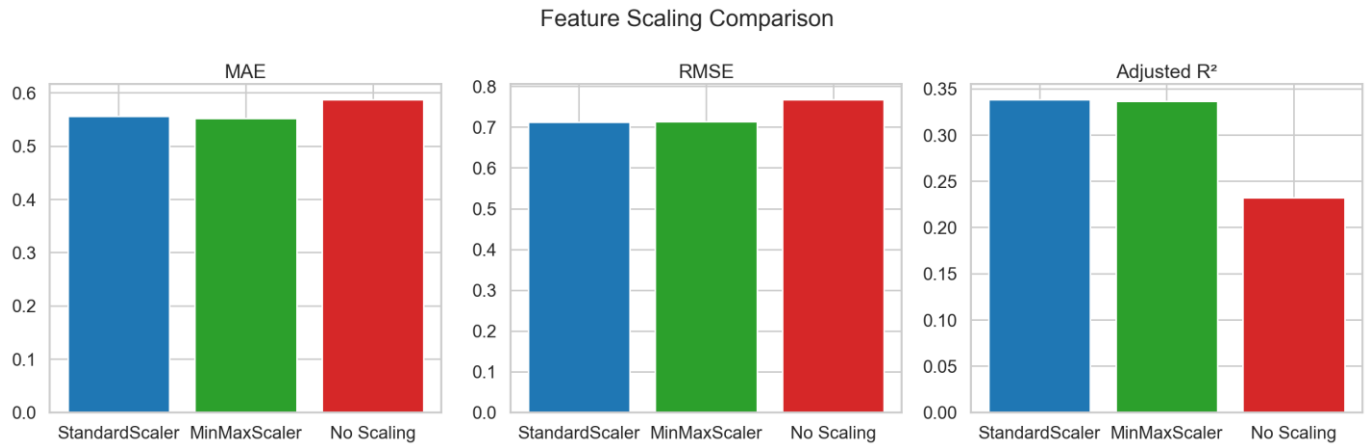
- Feature scaling improved both accuracy and model stability.
- StandardScaler provided the most effective preprocessing strategy.

Conclusion

- **Winner: StandardScaler**
- Best balance of:
  - Prediction Accuracy
  - Training Stability
  - Generalization







Feature Scaling Conclusion

Conclusion

- StandardScaler was selected for all subsequent analyses and became the foundation for the optimized deep learning pipeline.

Key Insights

- Scaling was one of the largest contributors to performance improvement.
- Standardization improved both accuracy and model stability.
- Neural networks benefited from features having consistent magnitudes.

Takeaway

Feature engineering and preprocessing decisions can influence performance as much as architectural decisions.

Activation Functions

What Is an Activation Function?

Activation functions introduce nonlinearity into neural networks, allowing them to learn complex relationships between inputs and outputs.

ReLU

- Fast computation
- Strong gradient propagation
- Most common deep learning activation

Sigmoid

- Produces outputs between 0 and 1
- Historically important in neural networks
- Can suffer from vanishing gradients

Tanh

- Produces outputs between -1 and 1
- Stronger gradients than Sigmoid
- Often provides smoother learning dynamics

Why Are They Important?

Without activation functions, neural networks would behave like simple linear models and lose much of their predictive power.

Activation Function Comparison

Objective

- Determine which activation function provides the strongest predictive performance and training behavior.

Why Compare?

- Activation functions influence:
  - Convergence speed
  - Stability
  - Predictive accuracy
  - Generalization performance

Evaluation Metrics

- MAE
- RMSE
- Adjusted R^2
- Learning Curves
  - Training Loss
  - Validation Loss

Key Question

- Which activation function is most effective for wine quality prediction?

Activation Function Results

Findings

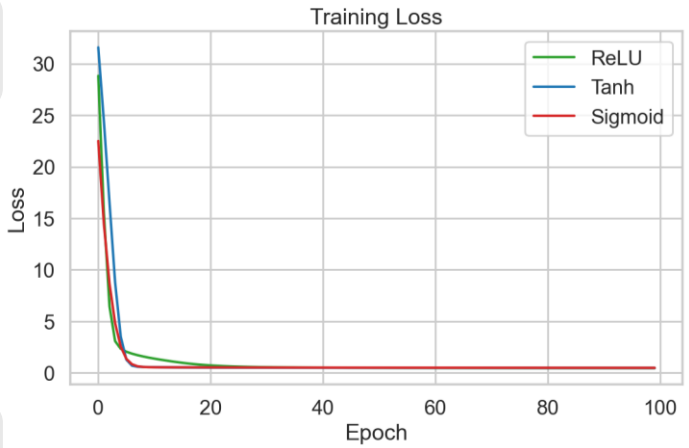
- ReLU converged the fastest.
- Sigmoid achieved the lowest MAE and RMSE and the highest Adjusted R^2.
- Tanh produced competitive but weaker results.
- All activation functions generalized effectively.

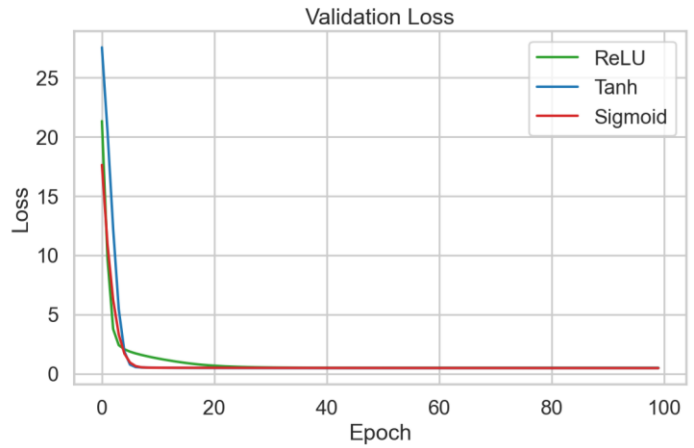
Interpretation

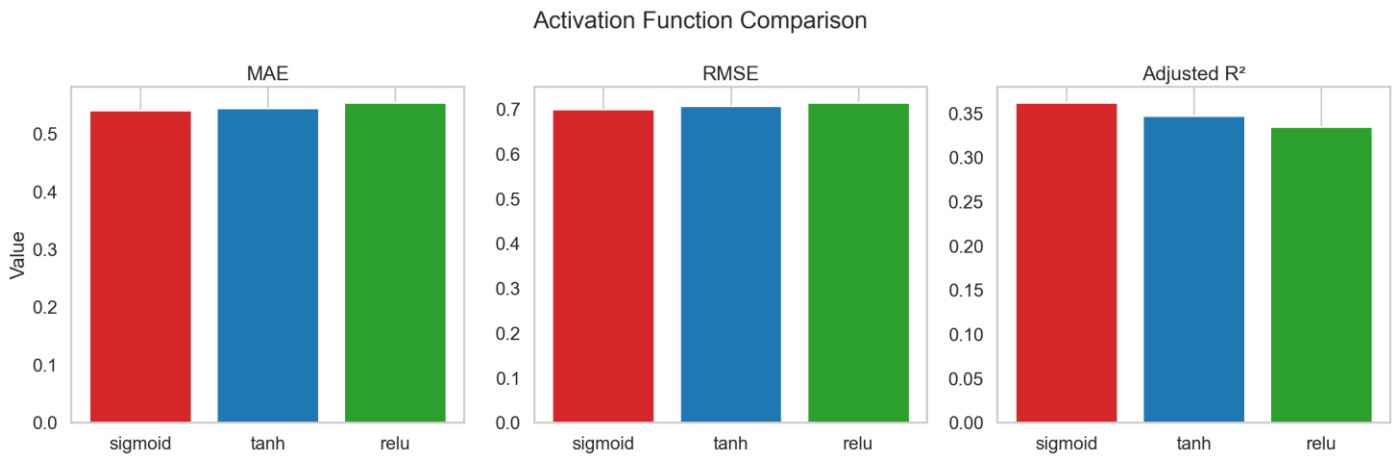
- Faster convergence did not guarantee better predictive performance.
- Sigmoid provided the strongest overall balance of accuracy and stability.

Conclusion

- **Winner: Sigmoid**
- Lowest error and highest Adjusted R^2.







Architecture Comparison

Objective

- Evaluate how network depth and complexity influence predictive performance.

Why Compare Architectures?

- The number of hidden layers and neurons determines a model's capacity to learn complex relationships.

Key Question

- How much network complexity is necessary to accurately model wine quality?

Architectures Evaluated

Evaluation Goal

Identify the architecture that achieves the strongest balance between:

- Accuracy
- Generalization
- Training efficiency
- Model complexity

→

Model A: 16

→

Model B: 32 → 16

↓

Model C: 64 → 32 → 16

→

Model D: 128 → 64 → 32 → 16

→

Why These Architectures?

The architectures were intentionally designed with increasing complexity to evaluate whether additional layers and neurons improve predictive performance.

Architecture Results

Findings

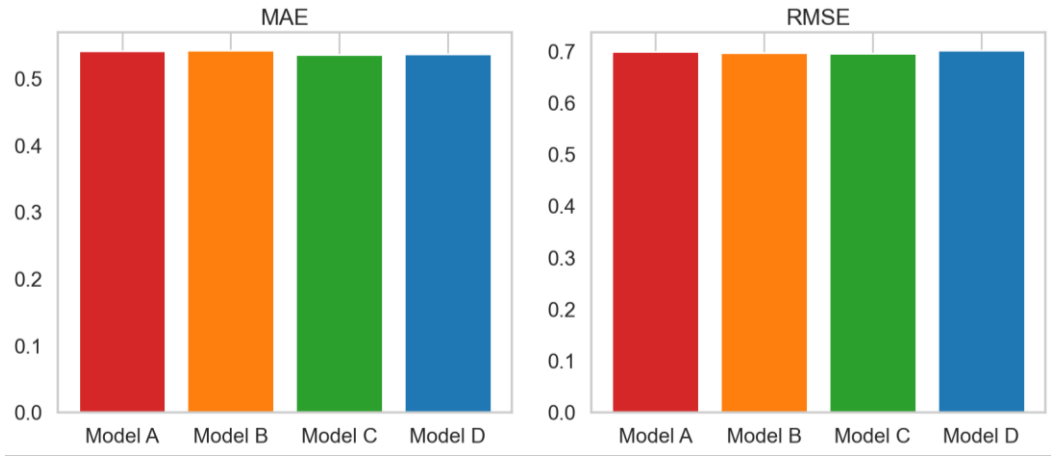
- Model C achieved the best overall performance.
- Model D increased training time without improving accuracy.
- Performance gains were modest across all architectures.
- Training time increased as model complexity increased.

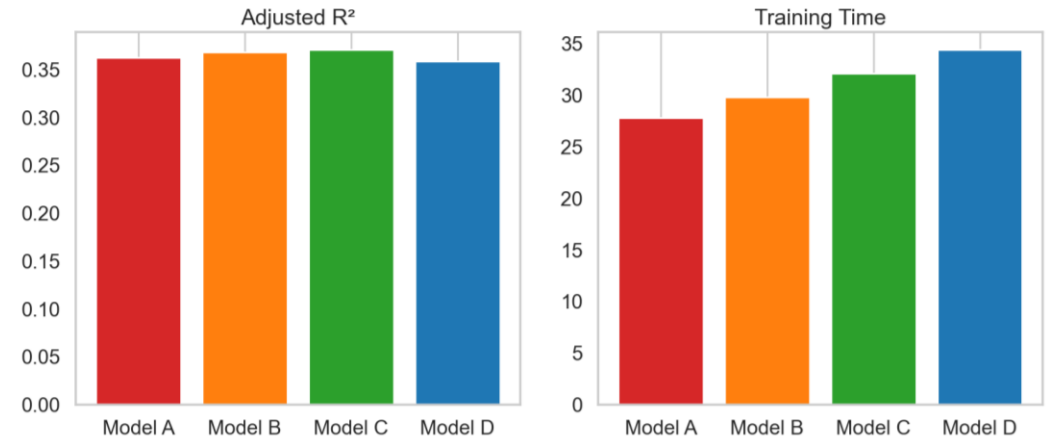
Interpretation

- Increasing complexity improved performance up to Model C.
- Additional layers beyond Model C produced diminishing returns.

Conclusion

- **Winner: Model C (64 → 32 → 16)**
- Strongest balance of performance and efficiency.





Epoch Optimization

Objective	What Is an Epoch?	Epoch Range Evaluated at:	Why Compare?	Key Question
• Determine the optimal number of training epochs that maximizes predictive performance while maintaining strong model generalization.	• One complete pass through the entire training dataset. • During each epoch, the neural network updates its weights to minimize prediction error. • Too few epochs may lead to underfitting. • Too many epochs may increase the risk of overfitting.	• 25, • 50, • 75, • 100, • 125, • 150, • 175, and • 200 epochs	• Training duration directly influences: • Model accuracy • Convergence behavior • Generalization performance • Computational cost	• What number of epochs provides the strongest balance between predictive accuracy and training efficiency?

Epoch Study Results

Findings

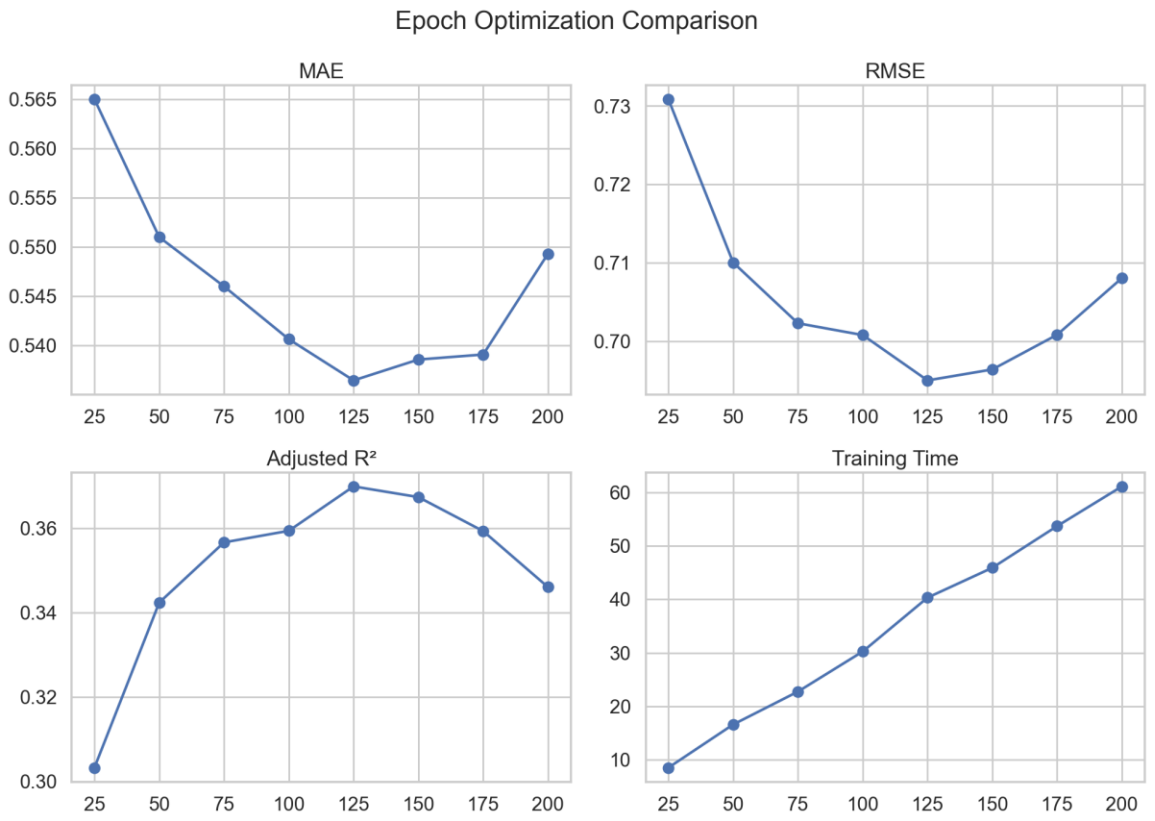
- Performance improved steadily from **25 to 125 epochs**.
- **125 epochs** achieved the lowest MAE and RMSE and the highest Adjusted R^2.
- Performance declined slightly beyond 125 epochs.
- Training time increased substantially with additional epochs.

Interpretation

- Additional epochs initially improved model learning.
- Gains diminished after **125 epochs**, while computational cost continued to increase.

Conclusion

- 🏆 **Winner: 125 Epochs**
- Achieved the lowest prediction error and highest Adjusted R^2 without the unnecessary training cost associated with longer training durations.



Optimizer Benchmarking

Objective

- Determine which optimization algorithm produces the strongest predictive performance while holding all other components constant.

SGD (Stochastic Gradient Descent)

- Traditional optimization method
- Simpler update mechanism
- Often requires additional tuning

RMSProp

- Adaptive learning-rate optimizer
- Designed for faster convergence
- Well-suited for neural networks

Adam

- Combines momentum and adaptive learning rates
- Widely used deep-learning optimizer
- Strong baseline performer

Why Optimizers Matter

- Optimizers control how neural network weights are updated during training.
- The choice of optimizer directly affects:
  - Convergence speed
  - Stability
  - Computational efficiency
  - Final model accuracy

Optimizer Benchmarking Results

Findings

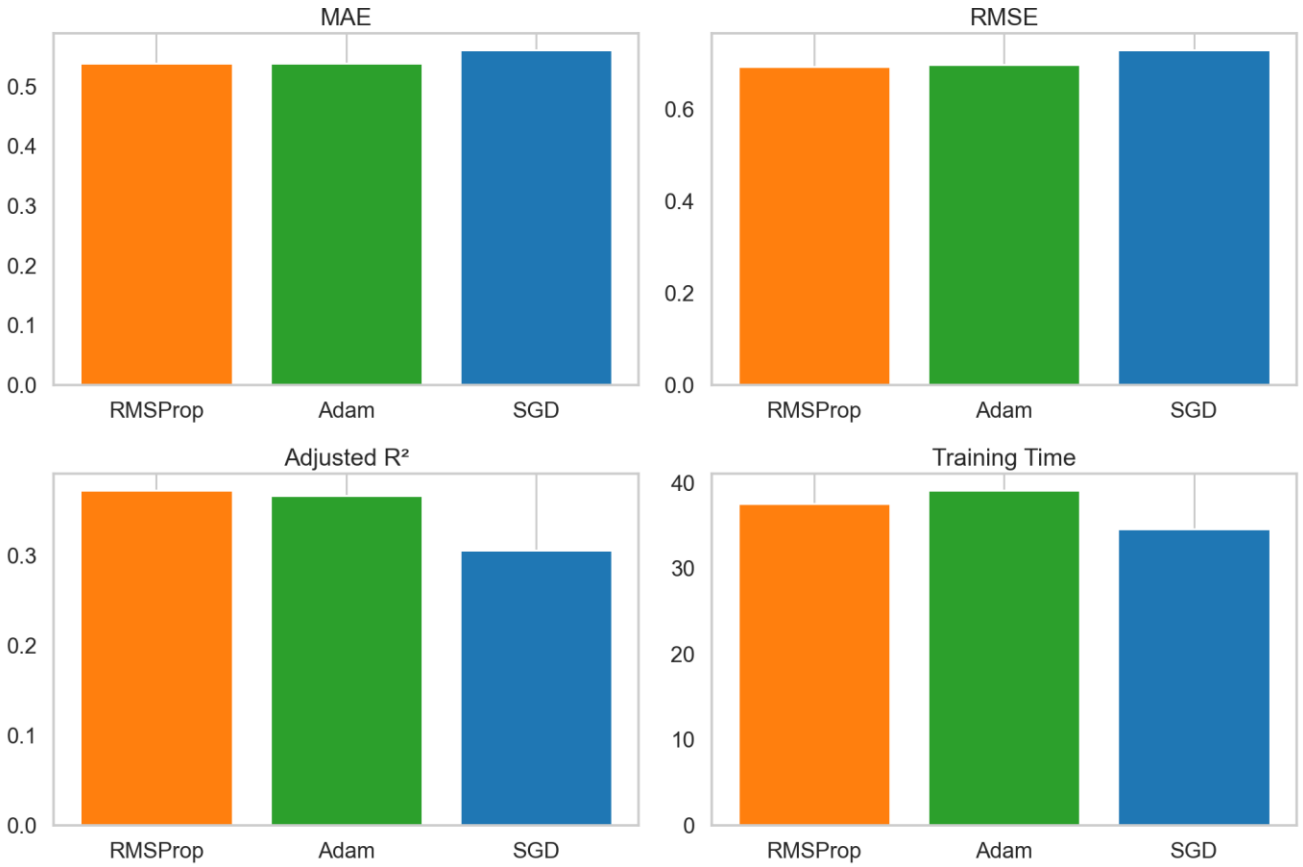
- RMSProp achieved the best overall performance.
- Adam produced similar results but did not outperform RMSProp.
- SGD had the highest error and lowest Adjusted R^2.
- Adaptive optimizers outperformed traditional gradient descent.

Interpretation

- Optimizer selection influenced model accuracy.
- RMSProp provided the strongest balance of performance and efficiency.

Conclusion

- **Winner: RMSProp**
- Best balance of:
  - Prediction Accuracy
  - Model Generalization
  - Training Efficiency



Regularization and Generalization

Objective

- Evaluate the impact of regularization techniques on predictive accuracy, model robustness, and generalization performance.

Why Regularization Matters

- Neural networks may memorize training data instead of learning general patterns.
- Regularization helps:
  - Reduce overfitting
  - Improve generalization
  - Increase model robustness
  - Stabilize training

Method	Purpose
Baseline	No regularization
L1	Encourages sparse weights
L2	Penalizes large weights
Dropout	Randomly disables neurons
Early Stopping	Stops training when improvement stalls

Regularization and Generalization Results

Findings

- Baseline achieved the best overall performance.
- Early Stopping produced competitive results.
- L1 and L2 reduced predictive performance.
- Additional regularization provided limited benefit.

Interpretation

- The model generalized effectively without regularization.
- Additional constraints reduced predictive accuracy.

Conclusion

- **Winner: Baseline (No Regularization)**
- Best balance of:
  - Prediction Accuracy
  - Generalization
  - Model Simplicity

Method	MAE	RMSE	Adjusted R^2
Baseline	0.5383	0.6933	0.3731
Early Stopping	0.5378	0.6941	0.3716
Dropout	0.5497	0.7043	0.3530
L2	0.5627	0.7299	0.3051
L1	0.6967	0.8796	-0.0092

Final Model Selection

Objective

- Identify the strongest-performing neural network configuration discovered throughout the optimization process.

Why Was This Model Selected?

- Best overall predictive performance.
- Highest Adjusted R^2.
- Lowest RMSE among tested configurations.
- Strong convergence behavior.
- Stable generalization performance.
- Fully reproducible experimental workflow.

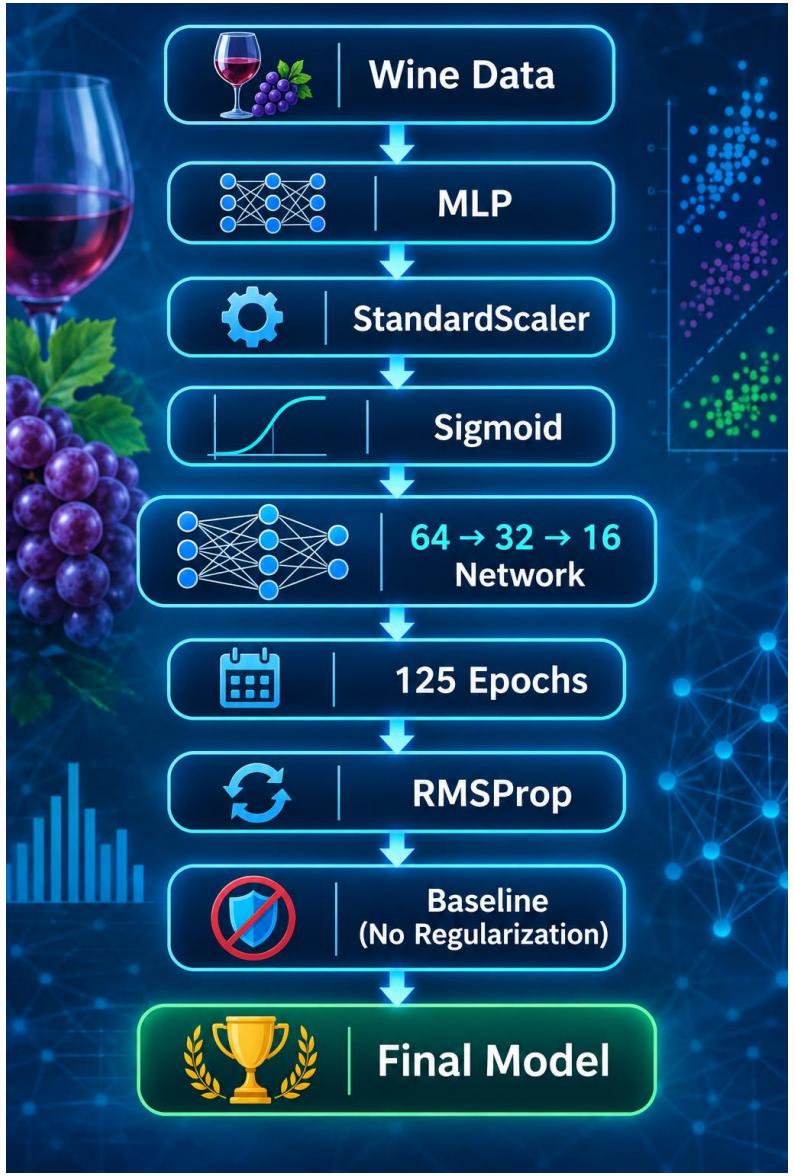
Final Configuration

- MLP
- StandardScaler
- Sigmoid
- Model C (64 → 32 → 16)
- RMSProp
- 125 Epochs
- No Regularization

Conclusion

- The final model combines the strongest-performing choices from every optimization stage.
- Together, these components produced the best balance of predictive accuracy, training stability, reproducibility, and model generalization.

Final Model Structure



Final Model Results

Final Performance Evaluation

- Final model outperformed the baseline across all evaluation metrics.
- Prediction accuracy improved while maintaining strong generalization.
- The model successfully captured relationships between wine chemistry and quality.

Key Findings

- Optimization improved overall predictive performance.
- Sigmoid and RMSProp delivered the strongest results.
- The final architecture consistently outperformed the baseline.
- No additional regularization was required.

Conclusion

- Optimized Model Selected
- Lower Prediction Error
- Higher Adjusted R^2
- Strong Generalization

Metric	Final Model
MAE	0.5383
RMSE	0.6933
Adjusted R^2	0.3731
Optimizer	RMSProp
Activation	Sigmoid
Architecture	64 → 32 → 16
Epochs	125

Performance Improvement

Findings

- MAE improved by **4.89%**
- RMSE improved by **5.87%**
- Adjusted R^2 improved by **27.60%**
- Every optimization stage contributed to performance gains.

Interpretation

- Incremental improvements accumulated throughout the optimization workflow.
- Preprocessing, architecture design, training strategy, and optimization choices collectively strengthened model performance.

Conclusion

- **Final Model Outperformed the Baseline MLP Model**
- Lower Prediction Error
- Higher Explanatory Power
- Better Overall Performance

Metric	Baseline MLP	Final Model
MAE	0.5660	0.5383
RMSE	0.7365	0.6933
Adjusted R^2	0.2924	0.3731

Dataset Limitations

Dataset Feature Limitations

- The Wine Quality dataset contains only physicochemical measurements and excludes several real-world factors that may influence wine quality, including:
  - Grape variety
  - Region of origin
  - Production methods
  - Storage conditions
  - Climate and harvest conditions
  - Expert tasting notes

Quality Score Limitations

- Wine quality scores are based on human sensory evaluations.
- Subjective scoring introduces variability between evaluators.
- Similar wines may receive different ratings.
- Some factors affecting quality may not be captured by chemical measurements alone.

Class Distribution Limitations

- Most wines received quality scores between 5 and 6.
- Very high-quality and very low-quality wines are underrepresented.
- Imbalanced target distributions can make prediction more challenging.

Key Takeaway

- Although the dataset provides valuable chemistry information, it does not fully capture all factors that influence wine quality. Additional variables could improve predictive performance and model interpretability.

Method Limitations

Architecture Search

- Only four architectures evaluated.
- Additional architectures may perform better.

Hyperparameters

- Limited hyperparameter tuning.
- Learning rates and batch sizes not fully explored.

Model Interpretability

- Neural networks function as black-box models.
- Predictions are difficult to explain.

Generalization

- Results are specific to the Wine Quality dataset.
- Performance may differ on other datasets.

Key Takeaway

- Best model identified from evaluated configurations.
- Additional optimization opportunities remain.

Next Steps

Hyperparameter Optimization

- Learning Rates
- Batch Sizes
- Weight Initialization
- Training Schedules

Architecture Exploration

- Deeper Networks
- Wider Layers
- Residual Networks
- Modern DL Architectures

Advanced Regularization

- Batch Normalization
- Learning-Rate Scheduling
- Improved Dropout
- Weight Decay

Explainable AI

- SHAP Values
- Feature Importance
- Model Interpretability

Ensemble Approaches

- Multiple Neural Networks
- Hybrid ML Approaches
- Ensemble Models

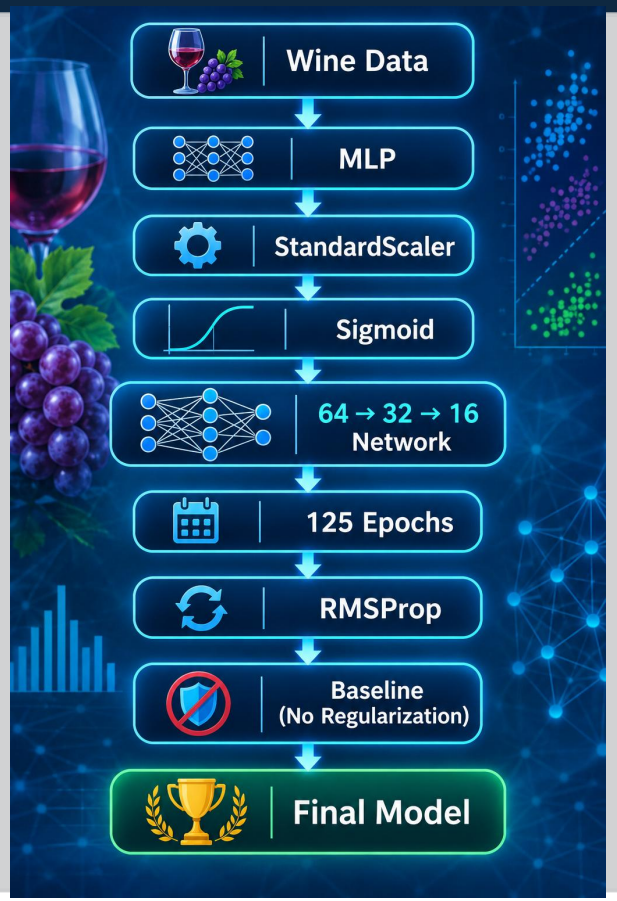
Long-Term Goal Develop a more accurate, robust, and explainable wine-quality prediction framework.

Project Conclusion

Key Findings

- StandardScaler improved performance.
- Sigmoid was the best activation function.
- Model C (64 → 32 → 16) achieved the best architecture balance.
- 125 Epochs optimized training.
- RMSProp was the strongest optimizer.
- No Regularization produced the best results.

Final Recommended Model



Final Takeaway

- MAE ↓ 4.89%
- RMSE ↓ 5.87%
- Adjusted R^2 ↑ 27.60%
- Systematic optimization transformed the baseline neural network into a stronger and more reliable predictive model.

Reference

Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). *Wine Quality Dataset*. UCI Machine Learning Repository. https://doi.org/10.24432/C56S3T